

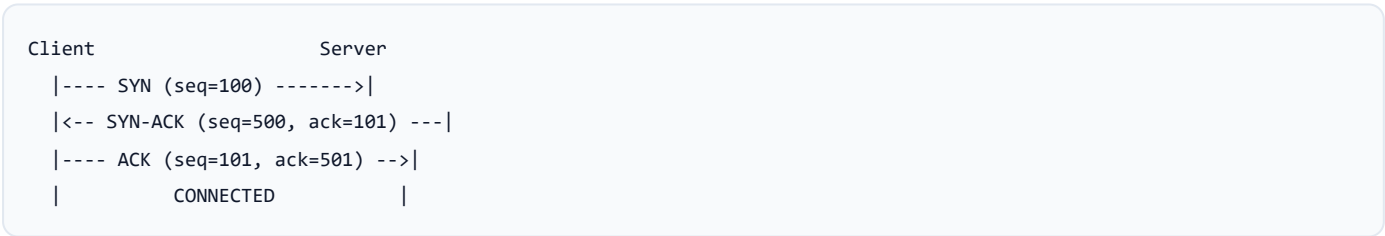
TCP, UDP & Ports in Depth

TCP Provides Order and Reliability

TCP (Transmission Control Protocol) is a reliable, ordered, byte-stream protocol built on IP. Key features:

- 3-way handshake: SYN → SYN-ACK → ACK
- Sequence numbers: keep segments ordered
- ACKs & retransmission: guarantee delivery
- Flow control: window sizing
- Congestion control: slow start, avoid collapse

The Three-Way Handshake



SYN → SYN-ACK → ACK. An attacker who can spoof this can hijack the connection – which requires predicting sequence numbers.

TCP Attacks

Attack	Mechanism
SYN flood	Send endless SYNs, never ACK → server memory exhausted (half-open)
Sequence prediction	Guess ACK sequence to inject data
Session hijacking	Steal a valid session via sniffing

Attack	Mechanism
FIN/RST injection	Resent reset to kill connections
TCP port scan	Try to complete handshakes on each port

UDP: Fire and Forget

UDP has no handshake, no ordering, no reliability. Faster, but:

- No connection state (spoofing trivial)

- No delivery guarantee

- Used by: DNS, NTP, DHCP, DHCPv6, QUIC/HTTP3, some video

UDP spoofing is trivial because there's no handshake to complete. An attacker forges a source IP, sends a request to an amplifier (e.g., a large DNS response), and the amplified response floods the victim – this is how **DNS amplification DDoS** works.

Ports: The Doors

A port is a 16-bit number (0-65535) identifying an application on a host. Common ones you MUST know:

Port	Service	Possible Attack
22	SSH	brute force, OpenSSH bugs
80	HTTP	web attacks
443	HTTPS	web attacks (TLS too)
445	SMB	EternalBlue/WannaCry
3389	RDP	brute force (FlawedAmmyy)
53	DNS	cache poisoning, reflection
21	FTP	cleartext creds
1433	MSSQL	brute force

Port Scanning Fundamentals

Open – port responded (reachable, service present)

Closed – reached but nothing listening (RST)

Filtered – packet dropped (firewall)

Scanners (Nmap) send probes and read the verdicts. The scan type (SYN, connect, ACK, FIN, UDP) changes which firewall behavior you can detect.

[← The OSI Model & TCP/IP Stack](#)

[IP Addressing, Subnetting & Ro...](#)